

FROM

SQL

TO

PANDAS



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SELECT ALL COLUMNS

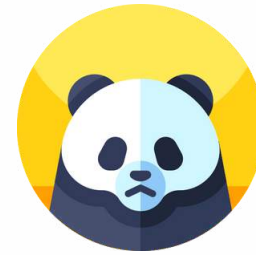


```
CREATE TABLE "df" (  
  "CustID" INTEGER,  
  "Name"    VARCHAR(10)  
);  
  
INSERT INTO "df" VALUES  
(1, 'Doe'),  
(2, 'Jo'),  
(3, 'Tod')  
;
```

```
SELECT *  
FROM df;
```

Output

CustID	Name
1	Doe
2	Jo
3	Tod



```
import pandas as pd  
  
data = {  
    'CustID': [1, 2, 3],  
    'Name': ['Doe', 'Jo', 'Tod']  
}  
  
df = pd.DataFrame(data)
```

```
print(df)
```

output

	CustID	Name
0	1	Doe
1	2	Jo
2	3	Tod



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SELECT SINGLE COLUMN



```
CREATE TABLE "df" (  
  "CustID" INTEGER,  
  "Name"    VARCHAR(10)  
);  
  
INSERT INTO "df" VALUES  
(1, 'Doe'),  
(2, 'Jo'),  
(3, 'Tod')  
;
```

```
SELECT "Name"  
FROM "df"
```

Name
Doe
Jo
Tod



```
import pandas as pd  
  
data = {  
    'CustID': [1, 2, 3],  
    'Name': ['Doe', 'Jo', 'Tod']  
}  
  
df = pd.DataFrame(data)
```

```
print(df['Name'])
```

```
0    Doe  
1     Jo  
2    Tod  
Name: Name, dtype: object
```

SELECT MULTIPLE COLUMNS



```
CREATE TABLE "df" (  
  "CustID" INTEGER,  
  "FirstName" VARCHAR,  
  "LastName" VARCHAR  
);  
  
INSERT INTO "df" VALUES  
(1, 'Doe', 'Pala'),  
(2, 'Jo', 'Noice'),  
(3, 'Tod', 'Palle')  
;  
  
SELECT  
  "CustID",  
  "FirstName"  
FROM "df";
```

...	CustID	FirstName
	1	Doe
	2	Jo
	3	Tod



```
import pandas as pd  
  
df = pd.DataFrame(  
    columns = [  
        'CustID',  
        'FirstName',  
        'LastName'  
    ]  
)  
  
df['CustID'] = [1, 2, 3]  
  
df['FirstName'] = ['Doe', 'Jo', 'Tod']  
  
df['LastName'] = ['Pala', 'Noice', 'Palle']  
  
print(df)
```

```
CustID  FirstName  LastName  
0      1      Doe      Pala  
1      2       Jo      Noice  
2      3      Tod      Palle
```

```
print(df[['CustID', 'FirstName']])
```

```
CustID  FirstName  
0      1      Doe  
1      2       Jo  
2      3      Tod
```



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FILTER ROWS

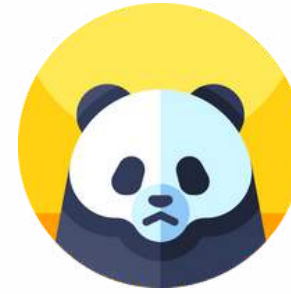


```
CREATE TABLE "df" (  
  "CustID" INTEGER,  
  "Name"    VARCHAR(10)  
);
```

```
INSERT INTO "df" VALUES  
(1, 'Doe'),  
(2, 'Jo'),  
(3, 'Tod')  
;
```

```
SELECT *  
  FROM "df"  
 WHERE "CustID" = '2';
```

CustID	Name	...
2	Jo	



```
import pandas as pd  
  
df = pd.DataFrame(  
    columns = ['CustID', 'Name']  
)  
  
df['CustID'] = [1, 2, 3]  
  
df['Name'] = ['Doe', 'Jo', 'Tod']
```

```
df[df['CustID'] == 2]
```

CustID	Name
2	Jo

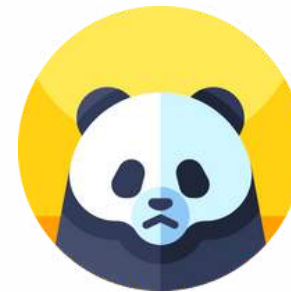
TOP N ROWS



```
CREATE TABLE "df" (  
  "CustID" INTEGER,  
  "Name"    VARCHAR(10)  
);  
  
INSERT INTO "df" VALUES  
(1, 'Doe'),  
(2, 'Jo'),  
(3, 'Tod')  
;
```

```
SELECT *  
  FROM "df"  
 LIMIT 1;
```

...	CustID	Name
	1	Doe



```
import pandas as pd  
  
df = pd.DataFrame(  
    columns = ['CustID', 'Name']  
)  
  
df['CustID'] = [1, 2, 3]  
  
df['Name'] = ['Doe', 'Jo', 'Tod']
```

```
print(df.head(1))
```

```
   CustID  Name  
0        1  Doe
```


DISTINCT VALUES

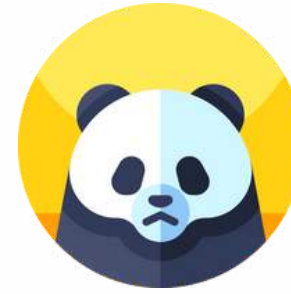


```
CREATE| TABLE "df" (  
  "CustID" INTEGER,  
  "Name"    VARCHAR  
);
```

```
INSERT INTO "df" VALUES  
(1, 'Doe'),  
(2, 'Jo'),  
(1, 'Tod')  
;
```

```
SELECT DISTINCT "CustID"  
FROM "df";
```

CustID
1
2



```
import pandas as pd  
  
df = pd.DataFrame(  
    columns = ['CustID', 'Name']  
)  
  
df['CustID'] = [1, 2, 1]  
  
df['Name'] = ['Doe', 'Jo', 'Tod']  
  
print(df)
```

```
CustID Name  
0      1  Doe  
1      2   Jo  
2      1  Tod
```

```
print(df.CustID.unique())
```

```
[1 2]
```


COUNT OF DISTINCT VALUES



```
create table "df" (  
  "CustID" INTEGER  
);
```

```
INSERT INTO df values  
(10),  
(20),  
(10);
```

```
SELECT COUNT(DISTINCT "CustID")  
FROM df;
```

...	COUNT(DISTINCT "CUSTID")
	2



```
import pandas as pd  
  
df = pd.DataFrame(  
    columns = ['CustID']  
)  
  
df['CustID'] = [10, 20, 10]  
  
print(df)
```

```
   CustID  
0       10  
1       20  
2       10
```

```
print(df.CustID.nunique())
```

```
2
```

COUNT OF TOTAL VALUES

table/dataframe

CustID	Name
10	Doe
20	Jo
30	Tod



```
SELECT COUNT(*) * (
  SELECT COUNT(*)
  FROM INFORMATION_SCHEMA.columns
  WHERE TABLE_CATALOG = 'DATABASE_NAME'
  AND TABLE_SCHEMA = 'SCHEMA_NAME'
  AND TABLE_NAME='df'
) AS "Size"
from "df";
```

Size
6



```
df.size
```

6



STRUCTURE OF TABLE

table/dataframe

CustID	Name
10	Doe
20	Jo
30	Tod



```
desc table "df";
```

name	...	type	kind
CustID		NUMBER(38,0)	COLUMN
Name		VARCHAR(20)	COLUMN



```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 3 entries, 0 to 2  
Data columns (total 2 columns):  
#   Column  Non-Null Count  Dtype  
---  -  
0   CustID   3 non-null      int64  
1   Name     3 non-null      object  
dtypes: int64(1), object(1)  
memory usage: 176.0+ bytes
```

DESCRIPTIVE STATISTICS



```
In [17]: df
```

```
Out[17]: 0    1
          1    2
          2    3
          3    4
          4    5
          Name: AGE, dtype: int64
```

```
In [21]: df.describe()
```

```
Out[21]: count    5.00000
          mean     3.00000
          std      1.58114
          min      1.00000
          25%      2.00000
          50%      3.00000
          75%      4.00000
          max       5.00000
          Name: AGE, dtype: float64
```



AGE
1
2
3
4
5

```
SELECT
    COUNT(age) AS "count"
    , AVG(age) AS "mean"
    , STDDEV(age) as "std"
    , MIN(age) as "min"
    , PERCENTILE_CONT(0.25) WITHIN GROUP (ORDER BY age) "25%"
    , PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY age) "50%"
    , PERCENTILE_CONT(0.75) WITHIN GROUP (ORDER BY age) "75%"
    , MAX(age) as "max"
FROM desc_stats;
```

...	count	mean	std	min	25%	50%	75%	max
	5	3.000000	1.58113883	1	2.000	3.000	4.000	5



GROUP BY SINGLE COLUMN



```
CREATE TABLE "df" (  
  "Gender" VARCHAR(1)  
  , "Population" INTEGER  
);  
  
INSERT INTO "df" VALUES  
( 'M' , 1),  
( 'F' , 1),  
( 'M' , 0),  
( 'F' , 1)  
;  
  
SELECT  
    "Gender"  
  , SUM("Population")  
FROM "df"  
GROUP BY "Gender"  
;
```

Gender	Population
M	1
F	2



```
: import pandas as pd  
  
df = {  
    'Gender': ['M', 'F', 'M', 'F'],  
    'Population': [1, 1, 0, 1]  
}  
  
df = pd.DataFrame(df)  
  
df
```

```
:  
  
    Gender  Population  
0        M           1  
1        F           1  
2        M           0  
3        F           1
```

```
: print(df.groupby('Gender').sum());
```

```
    Population  
Gender  
F           2  
M           1
```



SORT BY SINGLE COLUMN



```
create or replace table "df" (  
  "ID" INTEGER,  
  "Name" VARCHAR(10)  
);
```

```
INSERT INTO "df" values  
(5, 'Joe'),  
(2, 'Doe'),  
(4, 'Paula'),  
(3, 'John'),  
(1, 'Terry')  
;
```

```
SELECT *  
FROM "df"  
ORDER BY "ID";
```

...	ID	Name
	1	Terry
	2	Doe
	3	John
	4	Paula
	5	Joe



```
import pandas as pd
```

```
df = {  
    'ID': [5, 2, 4, 3, 1],  
    'NAME': ['Joe', 'Doe', 'Paula', 'John', 'Terry']  
}
```

```
df = pd.DataFrame(df)
```

```
df.sort_values(by=['ID'])
```

	ID	NAME
4	1	Terry
1	2	Doe
3	3	John
2	4	Paula
0	5	Joe



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SORT BY MULTIPLE COLUMNS



```
create or replace table "df" (  
  "ID"      INTEGER,  
  "Name"    VARCHAR(10),  
  "AGE"     INTEGER  
);
```

```
INSERT INTO "df" values  
(5, 'Joe', 20),  
(2, 'Doe', 50),  
(2, 'Paula', 10),  
(1, 'John', 40),  
(1, 'Terry', 30)  
;
```

```
SELECT *  
FROM "df"  
ORDER BY "ID", "AGE";
```

...	ID	Name	AGE
	1	Terry	30
	1	John	40
	2	Paula	10
	2	Doe	50
	5	Joe	20



```
import pandas as pd  
  
df = {  
    'ID': [5, 2, 2, 1, 1],  
    'NAME': ['Joe', 'Doe', 'Paula', 'John', 'Terry'],  
    'AGE': [20, 50, 10, 40, 30]  
}
```

```
df = pd.DataFrame(df)  
  
df.sort_values(by=['ID', 'AGE'])
```

	ID	NAME	AGE
4	1	Terry	30
3	1	John	40
2	2	Paula	10
1	2	Doe	50
0	5	Joe	20

**Sorted by default
in ascending order**



COUNT OF UNIQUE VALUES



```
create or replace table "df" (  
  "NAME"    VARCHAR(10)  
);
```

```
INSERT INTO "df" values  
( 'Joe' ),  
( 'Doe' ),  
( 'Paula' ),  
( 'Joe' ),  
( 'Doe' )  
;
```

```
SELECT "NAME", COUNT(*)  
FROM "df"  
GROUP BY "NAME"  
ORDER BY COUNT(*) DESC;
```

NAME	... COUNT(*)
Joe	2
Doe	2
Paula	1



```
import pandas as pd
```

```
df = ['Joe', 'Doe', 'Paula', 'Joe', 'Doe']
```

```
df = pd.DataFrame(df)
```

```
df.value_counts()
```

```
Doe      2  
Joe      2  
Paula    1  
Name: count, dtype: int64
```



DROP ROW - ALL COLUMNS DUPLICATED



```
create or replace table "df" (  
  "ID"      INTEGER,  
  "NAME"    VARCHAR(10)  
);
```

```
INSERT INTO "df" values  
(1, 'Joe'),  
(2, 'Jack'),  
(3, 'Paula'),  
(1, 'Joe')  
;
```

```
SELECT DISTINCT *  
FROM "df"  
;
```

ID	NAME
1	Joe
2	Jack
3	Paula



```
: import pandas as pd
```

```
: df = {  
    'ID': [1, 2, 3, 1],  
    'Name': ['Joe', 'Jack', 'Paul', 'Joe']  
}
```

```
: df = pd.DataFrame(df)
```

```
: df
```

	ID	Name
0	1	Joe
1	2	Jack
2	3	Paul
3	1	Joe

```
df.drop_duplicates()
```

	ID	Name
0	1	Joe
1	2	Jack
2	3	Paul



DROP ROW – KEY COLUMN DUPLICATED



```
create or replace table "df" (  
  "ID"      INTEGER,  
  "NAME"    VARCHAR(10)  
);  
  
INSERT INTO "df" values  
(1, 'Joe'),  
(2, 'Jack'),  
(3, 'Paula'),  
(1, 'Doe')  
;  
  
DELETE FROM "df" T1  
USING  
(  
  SELECT  
    ID,  
    NAME  
  FROM "df"  
  QUALIFY ROW_NUMBER() OVER (PARTITION BY ID ORDER BY ID ASC) = '2'  
) T2  
WHERE T1."ID" = T2."ID" AND T1."NAME" = T2."NAME"  
;  
  
SELECT *  
FROM "df"  
;
```

...	ID	NAME
	1	Joe
	2	Jack
	3	Paula



```
import pandas as pd
```

```
df = {  
    'ID': [1, 2, 3, 1],  
    'Name': ['Joe', 'Jack', 'Paul', 'Doe']  
}
```

```
df = pd.DataFrame(df)
```

```
df
```

	ID	Name
0	1	Joe
1	2	Jack
2	3	Paul
3	1	Doe

```
df.drop_duplicates(subset=['ID'])
```

	ID	Name
0	1	Joe
1	2	Jack
2	3	Paul



INNER JOIN



```
create or replace table "df1" (  
  "ID"      INTEGER,  
  "NAME"    VARCHAR(10)  
);  
INSERT INTO "df1" values  
(1, 'Joe'),  
(2, 'Jack'),  
(3, 'Paula');  
create or replace table "df2" (  
  "ID"      INTEGER,  
  "AGE"     INTEGER  
);  
INSERT INTO "df2" values  
(1, 10),  
(2, 20),  
(4, 40);  
SELECT "df1".ID, NAME, AGE  
FROM "df1"  
INNER JOIN "df2"  
ON "df1".ID = "df2".ID;
```

...	ID	NAME	AGE
	1	Joe	10
	2	Jack	20



```
import pandas as pd
```

```
df1 = {  
    'ID': [1, 2, 3],  
    'Name': ['Joe', 'Jack', 'Paula']  
}  
  
df1 = pd.DataFrame(df1)  
  
df2 = {  
    'ID': [1, 2, 4],  
    'Age': [10, 20, 40],  
}  
  
df2 = pd.DataFrame(df2)
```

```
df = pd.merge(df1, df2, on='ID', how='inner')  
  
df
```

	ID	Name	Age
0	1	Joe	10
1	2	Jack	20



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LEFT JOIN



```
create or replace table "df1" (  
  "ID"      INTEGER,  
  "NAME"    VARCHAR(10)  
);  
INSERT INTO "df1" values  
(1, 'Joe'),  
(2, 'Jack'),  
(3, 'Paula');  
create or replace table "df2" (  
  "ID"      INTEGER,  
  "AGE"     INTEGER  
);  
INSERT INTO "df2" values  
(1, 10),  
(2, 20),  
(4, 40);  
SELECT "df1".ID, NAME, AGE  
FROM "df1"  
LEFT JOIN "df2"  
ON "df1".ID = "df2".ID;
```

ID	NAME	AGE
1	Joe	10
2	Jack	20
3	Paula	null



```
import pandas as pd
```

```
df1 = {  
    'ID': [1, 2, 3],  
    'Name': ['Joe', 'Jack', 'Paula']  
}
```

```
df1 = pd.DataFrame(df1)
```

```
df2 = {  
    'ID': [1, 2, 4],  
    'Age': [10, 20, 40],  
}
```

```
df2 = pd.DataFrame(df2)
```

```
df = pd.merge(df1, df2, on='ID', how='left')
```

```
df
```

	ID	Name	Age
0	1	Joe	10.0
1	2	Jack	20.0
2	3	Paula	NaN

RIGHT JOIN



```
create or replace table "df1" (  
  "ID"      INTEGER,  
  "NAME"    VARCHAR(10)  
);  
INSERT INTO "df1" values  
(1, 'Joe'),  
(2, 'Jack'),  
(3, 'Paula');  
create or replace table "df2" (  
  "ID"      INTEGER,  
  "AGE"     INTEGER  
);  
INSERT INTO "df2" values  
(1, 10),  
(2, 20),  
(4, 40);  
SELECT "df2".ID, NAME, AGE  
FROM "df1"  
RIGHT JOIN "df2"  
ON "df1".ID = "df2".ID;
```

ID	NAME	...	AGE
1	Joe		10
2	Jack		20
4	null		40



```
: import pandas as pd
```

```
: df1 = {  
    'ID': [1, 2, 3],  
    'Name': ['Joe', 'Jack', 'Paula']  
}
```

```
df1 = pd.DataFrame(df1)
```

```
df2 = {  
    'ID': [1, 2, 4],  
    'Age': [10, 20, 40],  
}
```

```
df2 = pd.DataFrame(df2)
```

```
: df = pd.merge(df1, df2, on='ID', how='right')
```

```
df
```

```
:
```

	ID	Name	Age
0	1	Joe	10
1	2	Jack	20
2	4	NaN	40



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FULL JOIN



```
create or replace table "df1" (  
  "ID"      INTEGER,  
  "NAME"    VARCHAR(10)  
);  
INSERT INTO "df1" values  
(1, 'Joe'),  
(2, 'Jack'),  
(3, 'Paula');  
create or replace table "df2" (  
  "ID"      INTEGER,  
  "AGE"     INTEGER  
);  
INSERT INTO "df2" values  
(1, 10),  
(2, 20),  
(4, 40);  
SELECT COALESCE("df1".ID, "df2".ID) AS ID,  
       NAME,  
       AGE  
FROM "df1"  
FULL JOIN "df2"  
ON "df1".ID = "df2".ID;  
ORDER BY 1;
```

ID	NAME	AGE
1	Joe	10
2	Jack	20
3	Paula	null
4	null	40



```
: import pandas as pd  
  
: df1 = {  
    'ID': [1, 2, 3],  
    'Name': ['Joe', 'Jack', 'Paula']  
}  
df1 = pd.DataFrame(df1)  
  
df2 = {  
    'ID': [1, 2, 4],  
    'Age': [10, 20, 40],  
}  
df2 = pd.DataFrame(df2)  
  
: df = pd.merge(df1, df2, on='ID', how='outer')  
df
```

```
:  
      ID  Name  Age  
0      1   Joe  10.0  
1      2  Jack  20.0  
2      3  Paula  NaN  
3      4   NaN  40.0
```



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CROSS JOIN



```
create or replace table "df1" (  
  "ID"      INTEGER,  
  "NAME"    VARCHAR(10)  
);  
INSERT INTO "df1" values  
(1, 'Joe'),  
(2, 'Jack');  
create or replace table "df2" (  
  "ID"      INTEGER,  
  "AGE"     INTEGER  
);  
INSERT INTO "df2" values  
(1, 10),  
(2, 20);  
SELECT *  
FROM "df1"  
CROSS JOIN "df2"  
;
```

...	ID	NAME	ID_2	AGE
	1	Joe	1	10
	1	Joe	2	20
	2	Jack	1	10
	2	Jack	2	20



```
import pandas as pd
```

```
df1 = {  
    'ID': [1, 2],  
    'Name': ['Joe', 'Jack']  
}
```

```
df1 = pd.DataFrame(df1)
```

```
df2 = {  
    'ID': ['1', '2'],  
    'AGE': [10, 20]  
}
```

```
df2 = pd.DataFrame(df2)
```

```
df = pd.merge(df1, df2, how='cross')
```

```
df
```

	ID_x	Name	ID_y	AGE
0	1	Joe	1	10
1	1	Joe	2	20
2	2	Jack	1	10
3	2	Jack	2	20



UNION ALL BY SINGLE COLUMN



```
CREATE OR REPLACE TABLE "df1" (  
  "ID"      INTEGER,  
  "NAME"    VARCHAR(10)  
);  
INSERT INTO "df1" values  
(1, 'Joe'),  
(2, 'Jack');  
--  
CREATE OR REPLACE TABLE "df2" (  
  "ID"      INTEGER,  
  "NAME"    VARCHAR(10)  
);  
INSERT INTO "df2" values  
(1, 'Joe'),  
(4, 'Doe');  
--  
SELECT NAME  
FROM "df1"  
UNION ALL  
SELECT NAME  
FROM "df2"
```

NAME
Joe
Jack
Joe
Doe



```
import pandas as pd
```

```
df1 = {  
    'ID': [1, 2],  
    'Name': ['Joe', 'Jack']  
}  
  
df1 = pd.DataFrame(df1)  
  
df2 = {  
    'ID': [1, 4],  
    'Name': ['Joe', 'Doe']  
}  
  
df2 = pd.DataFrame(df2)
```

```
df = pd.concat([df1['Name'], df2['Name']], ignore_index=True)  
  
# ignore_index=True will reindex the dataframe  
  
df
```

```
0      Joe  
1     Jack  
2      Joe  
3      Doe
```



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UNION ALL BY ALL COLUMNS



```
CREATE OR REPLACE TABLE "df1" (  
  "ID"      INTEGER,  
  "NAME"    VARCHAR(10)  
);  
INSERT INTO "df1" values  
(1, 'Joe'),  
(2, 'Jack');  
--  
CREATE OR REPLACE TABLE "df2" (  
  "ID"      INTEGER,  
  "NAME"    VARCHAR(10)  
);  
INSERT INTO "df2" values  
(1, 'Joe'),  
(4, 'Doe');  
--  
SELECT *  
FROM "df1"  
UNION ALL  
SELECT *  
FROM "df2";
```

ID	NAME
1	Joe
2	Jack
1	Joe
4	Doe



```
import pandas as pd
```

```
df1 = {  
    'ID': [1, 2],  
    'Name': ['Joe', 'Jack']  
}
```

```
df1 = pd.DataFrame(df1)
```

```
df2 = {  
    'ID': [1, 4],  
    'Name': ['Joe', 'Doe']  
}
```

```
df2 = pd.DataFrame(df2)
```

```
df = pd.concat([df1, df2], ignore_index=True)  
  
# ignore_index=True will reindex the dataframe  
  
df
```

	ID	Name
0	1	Joe
1	2	Jack
2	1	Joe
3	4	Doe



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UNION BY SINGLE COLUMN



```
CREATE OR REPLACE TABLE "df1" (  
  "ID"      INTEGER,  
  "NAME"    VARCHAR(10)  
);  
INSERT INTO "df1" values  
(1, 'Joe'),  
(2, 'Jack');  
--  
CREATE OR REPLACE TABLE "df2" (  
  "ID"      INTEGER,  
  "NAME"    VARCHAR(10)  
);  
INSERT INTO "df2" values  
(1, 'Joe'),  
(4, 'Doe');  
--  
SELECT NAME  
FROM "df1"  
UNION  
SELECT NAME  
FROM "df2"  
;
```

NAME ...
Joe
Jack
Doe



```
import pandas as pd
```

```
df1 = {  
    'ID': [1, 2],  
    'Name': ['Joe', 'Jack']  
}
```

```
df1 = pd.DataFrame(df1)
```

```
df2 = {  
    'ID': [1, 4],  
    'Name': ['Joe', 'Doe']  
}
```

```
df2 = pd.DataFrame(df2)
```

```
df = pd.concat([df1['Name'], df2['Name']], ignore_index=True).drop_duplicates()
```

```
# ignore_index=True will reindex the dataframe
```

```
df
```

```
0      Joe  
1     Jack  
3      Doe
```



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UNION BY ALL COLUMNS



```
CREATE OR REPLACE TABLE "df1" (  
  "ID"      INTEGER,  
  "NAME"    VARCHAR(10)  
);  
INSERT INTO "df1" values  
(1, 'Joe'),  
(2, 'Jack');  
--  
CREATE OR REPLACE TABLE "df2" (  
  "ID"      INTEGER,  
  "NAME"    VARCHAR(10)  
);  
INSERT INTO "df2" values  
(1, 'Joe'),  
(4, 'Doe');  
--  
SELECT *  
FROM "df1"  
UNION  
SELECT *  
FROM "df2";
```

ID	NAME
1	Joe
2	Jack
4	Doe



```
import pandas as pd
```

```
df1 = {  
    'ID': [1, 2],  
    'Name': ['Joe', 'Jack']  
}  
  
df1 = pd.DataFrame(df1)  
  
df2 = {  
    'ID': [1, 4],  
    'Name': ['Joe', 'Doe']  
}  
  
df2 = pd.DataFrame(df2)
```

```
df = pd.concat([df1, df2], ignore_index=True).drop_duplicates()  
df
```

	ID	Name
0	1	Joe
1	2	Jack
3	4	Doe



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ADD SINGLE COLUMN



```
CREATE TABLE "data" (  
  "ID" INTEGER  
,  
  "NAME" VARCHAR(10)  
);
```

```
INSERT INTO "data" VALUES  
(1, 'Joe'),  
(2, 'Jack'),  
(3, 'Paula')  
;
```

```
ALTER TABLE "data" ADD COLUMN AGE INTEGER;
```

```
UPDATE "data" SET AGE = 10 WHERE ID = '1';  
UPDATE "data" SET AGE = 20 WHERE ID = '2';  
UPDATE "data" SET AGE = 40 WHERE ID = '3';
```

```
SELECT * FROM "data";
```

...	ID	NAME	AGE
	1	Joe	10
	2	Jack	20
	3	Paula	40



```
data = {  
  'ID': [1, 2, 3],  
  'Name': ['Joe', 'Jack', 'Paula']  
}
```

```
df = pd.DataFrame(data)
```

```
df
```

	ID	Name
0	1	Joe
1	2	Jack
2	3	Paula

```
df['Age'] = [10, 20, 40]
```

```
df
```

	ID	Name	Age
0	1	Joe	10
1	2	Jack	20
2	3	Paula	40

DROP SINGLE COLUMN



```
CREATE or replace TABLE "data" (  
  "ID" INTEGER  
,  
  "NAME" VARCHAR(10)  
,  
  "AGE" INTEGER  
);  
  
INSERT INTO "data" VALUES  
(1, 'Joe', '10'),  
(2, 'Jack', '20'),  
(3, 'Paula', '40')  
;  
  
ALTER TABLE "data" DROP COLUMN "AGE";  
  
SELECT * FROM "data";
```

...	ID	NAME
	1	Joe
	2	Jack
	3	Paula



```
data = {  
  'ID': [1, 2, 3],  
  'Name': ['Joe', 'Jack', 'Paula']  
  'Age': [10, 20, 40]  
}
```

```
df = pd.DataFrame(data)
```

df

	ID	Name	Age
0	1	Joe	10
1	2	Jack	20
2	3	Paula	40

```
df.drop(['Age'], axis=1, inplace=True)
```

df

	ID	Name
0	1	Joe
1	2	Jack
2	3	Paula



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DROP MULTIPLE COLUMNS



```
CREATE or replace TABLE "data" (  
  "ID" INTEGER  
,  
  "NAME" VARCHAR(10)  
,  
  "AGE" INTEGER  
);
```

```
INSERT INTO "data" VALUES  
(1, 'Joe', '10'),  
(2, 'Jack', '20'),  
(3, 'Paula', '40')  
;
```

```
ALTER TABLE "data"  
DROP COLUMN "AGE", "NAME";
```

```
SELECT * FROM "data";
```

ID
1
2
3



```
: import pandas as pd
```

```
: data = {  
    'ID': [1, 2, 3],  
    'Name': ['Joe', 'Jack', 'Paula'],  
    'Age': [10, 20, 40]  
}
```

```
: df = pd.DataFrame(data)
```

```
: df
```

```
:
```

	ID	Name	Age
0	1	Joe	10
1	2	Jack	20
2	3	Paula	40

```
: df.drop(['Age', 'Name'], axis=1, inplace=True)
```

```
: df
```

```
:
```

	ID
0	1
1	2
2	3



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RENAME SINGLE COLUMN



```
CREATE TABLE "data" (  
  "ID" INTEGER  
,  
  "NAME" VARCHAR(10)  
,  
  "AGE" INTEGER  
,  
  "HEIHT" VARCHAR(10)  
);  
  
ALTER TABLE "data"  
RENAME COLUMN "HEIHT" to "HEIGHT";  
  
DESC TABLE "data";
```

name	type
ID	NUMBER(38,0)
NAME	VARCHAR(10)
AGE	NUMBER(38,0)
HEIGHT	VARCHAR(10)



```
: import pandas as pd
```

```
: data = {  
    'ID': [1, 2, 3],  
    'Name': ['Joe', 'Jack', 'Paula'],  
    'Age': [10, 20, 40],  
    'Heiht': ['1.65', '1.78', '1.82']  
}
```

```
: df = pd.DataFrame(data)
```

```
: df.rename(columns = {'Heiht': 'Height'}, inplace = True)
```

```
: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 3 entries, 0 to 2  
Data columns (total 4 columns):  
#   Column    Non-Null Count  Dtype  
---  ---  
0   ID        3 non-null      int64  
1   Name      3 non-null      object  
2   Age       3 non-null      int64  
3   Height    3 non-null      object  
dtypes: int64(2), object(2)  
memory usage: 228.0+ bytes
```

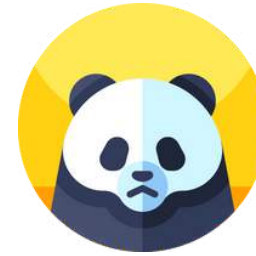


RENAME MULTIPLE COLUMNS



```
CREATE TABLE "data" (  
    "ID" INTEGER  
,  
    "NME" VARCHAR(10)  
,  
    "AGE" INTEGER  
,  
    "HEIHT" VARCHAR(10)  
);  
  
ALTER TABLE "data"  
RENAME COLUMN "NME" to "NAME";  
  
ALTER TABLE "data"  
RENAME COLUMN "HEIHT" to "HEIGHT";  
  
DESC TABLE "data";
```

name	type
ID	NUMBER(38,0)
NAME	VARCHAR(10)
AGE	NUMBER(38,0)
HEIGHT	VARCHAR(10)



```
import pandas as pd
```

```
data = {  
    'ID': [1, 2, 3],  
    'Nme': ['Joe', 'Jack', 'Paula'],  
    'Age': [10, 20, 40],  
    'Heiht': ['1.65', '1.78', '1.82']  
}
```

```
df = pd.DataFrame(data)
```

```
df.rename(columns = {'Nme': 'Name',  
                    'Heiht': 'Height'},  
         inplace = True)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 3 entries, 0 to 2  
Data columns (total 4 columns):  
#   Column  Non-Null Count  Dtype  
---  -  
0   ID      3 non-null       int64  
1   Name    3 non-null       object  
2   Age     3 non-null       int64  
3   Height  3 non-null       object  
dtypes: int64(2), object(2)  
memory usage: 228.0+ bytes
```


UPDATE SINGLE COLUMN SINGLE ROW



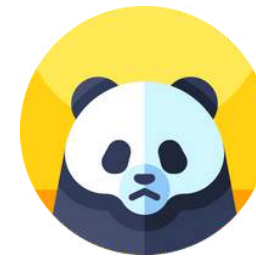
```
CREATE OR REPLACE TABLE "data" (  
  "ID" INTEGER  
,  
  "NAME" VARCHAR(10)  
,  
  "AGE" INTEGER  
);
```

```
INSERT INTO "data" VALUES  
(1, 'Joe', 10),  
(2, 'Jack', 20),  
(3, 'Paula', 30)  
;
```

```
UPDATE "data"  
SET "NAME" = 'John'  
WHERE "ID" = '3';
```

```
SELECT * FROM "data";
```

ID	NAME ...	AGE
1	Joe	10
2	Jack	20
3	John	30



```
: import pandas as pd
```

```
: data = {  
    'ID': [1, 2, 3],  
    'Name': ['Joe', 'Jack', 'Paula'],  
    'Age': [10, 20, 30]  
}
```

```
df = pd.DataFrame(data)
```

```
df
```

```
:
```

	ID	Name	Age
0	1	Joe	10
1	2	Jack	20
2	3	Paula	30

```
: df.loc[df['ID'] == 3, 'Name'] = ['John']
```

```
df
```

```
:
```

	ID	Name	Age
0	1	Joe	10
1	2	Jack	20
2	3	John	30



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UPDATE MULTIPLE COLUMNS SINGLE ROW



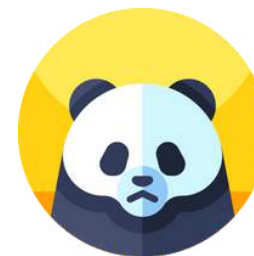
```
CREATE OR REPLACE TABLE "data" (  
  "ID" INTEGER  
,  
  "NAME" VARCHAR(10)  
,  
  "AGE" INTEGER  
);
```

```
INSERT INTO "data" VALUES  
(1, 'Joe', 10),  
(2, 'Jack', 20),  
(3, 'Paula', 40)  
;
```

```
UPDATE "data"  
SET "NAME" = 'John', "AGE" = '30'  
WHERE "ID" = '3';
```

```
SELECT * FROM "data";
```

...	ID	NAME	AGE
	1	Joe	10
	2	Jack	20
	3	John	30



```
import pandas as pd
```

```
data = {  
    'ID': [1, 2, 3],  
    'Name': ['Joe', 'Jack', 'Paula'],  
    'Age': [10, 20, 40]  
}  
  
df = pd.DataFrame(data)  
  
df
```

	ID	Name	Age
0	1	Joe	10
1	2	Jack	20
2	3	Paula	40

```
df.loc[df['ID'] == 3, ['Name', 'Age']] = ['John', 30]  
  
df
```

	ID	Name	Age
0	1	Joe	10
1	2	Jack	20
2	3	John	30



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UPDATE SINGLE COLUMN MULTIPLE ROWS



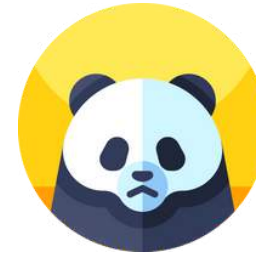
```
CREATE OR REPLACE TABLE "data" (  
  "ID" INTEGER  
,  
  "NAME" VARCHAR(10)  
,  
  "AGE" INTEGER  
);
```

```
INSERT INTO "data" VALUES  
(1, 'Joe', 10),  
(2, 'Jack', 200),  
(3, 'Paula', 400);
```

```
UPDATE "data"  
SET "AGE" = '99'  
WHERE "AGE" > '100';
```

```
SELECT * FROM "data";
```

...	ID	NAME	AGE
	1	Joe	10
	2	Jack	99
	3	Paula	99



```
import pandas as pd
```

```
data = {  
    'ID': [1, 2, 3],  
    'Name': ['Joe', 'Jack', 'Paula'],  
    'Age': [10, 200, 400]  
}  
  
df = pd.DataFrame(data)  
  
df
```

	ID	Name	Age
0	1	Joe	10
1	2	Jack	200
2	3	Paula	400

```
df.loc[df['Age'] > 100, 'Age'] = 99  
  
df
```

	ID	Name	Age
0	1	Joe	10
1	2	Jack	99
2	3	Paula	99



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UPDATE MULTIPLE COLUMNS MULTIPLE ROWS



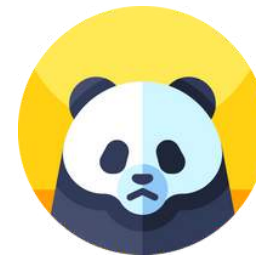
```
CREATE OR REPLACE TABLE "data" (  
  "ID" INTEGER  
,  
  "NAME" VARCHAR(10)  
,  
  "AGE" INTEGER  
);
```

```
INSERT INTO "data" VALUES  
(1, 'Joe', 10),  
(2, 'Jack', 200),  
(3, 'Paula', 400);
```

```
UPDATE "data"  
SET "AGE" = '99', "NAME" = 'John'  
WHERE "AGE" > '100';
```

```
SELECT * FROM "data";
```

ID	NAME	AGE
1	Joe	10
2	John	99
3	John	99



```
import pandas as pd
```

```
data = {  
  'ID': [1, 2, 3],  
  'Name': ['Joe', 'Jack', 'Paula'],  
  'Age': [10, 200, 400]  
}
```

```
df = pd.DataFrame(data)
```

```
df
```

	ID	Name	Age
0	1	Joe	10
1	2	Jack	200
2	3	Paula	400

```
df.loc[df['Age'] > 100, ['Name', 'Age']] = ['John', 99]  
df
```

	ID	Name	Age
0	1	Joe	10
1	2	John	99
2	3	John	99

UPDATE SINGLE COLUMN ALL ROWS



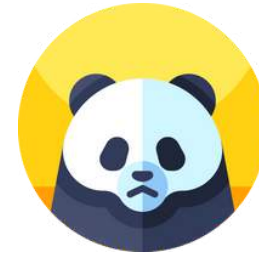
```
CREATE OR REPLACE TABLE "data" (  
  "ID" INTEGER  
,  
  "NAME" VARCHAR(10)  
,  
  "AGE" INTEGER  
);
```

```
INSERT INTO "data" VALUES  
(1, 'Joe', 10),  
(2, 'Jack', 20),  
(3, 'Paula', 30)  
;
```

```
UPDATE "data"  
SET "AGE" = 99;
```

```
SELECT * FROM "data";
```

...	ID	NAME	AGE
	1	Joe	99
	2	Jack	99
	3	Paula	99



```
data = {  
  'ID': [1, 2, 3],  
  'Name': ['Joe', 'Jack', 'Paula'],  
  'Age': [10, 20, 30]  
}
```

```
df = pd.DataFrame(data)
```

```
df
```

	ID	Name	Age
0	1	Joe	10
1	2	Jack	20
2	3	Paula	30

```
df['Age'] = 99
```

```
df
```

	ID	Name	Age
0	1	Joe	99
1	2	Jack	99
2	3	Paula	99



That's a wrap!

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